Day 14: Advanced Load Balancing & Networking

In a mature microservices environment, simple traffic forwarding is insufficient. You require granular control over how different versions of a service are accessed, how internal traffic is isolated from the public internet, and how to protect the network perimeter.

Section 1: Multi-Service Routing

Multi-service routing (or Content-Based Routing) allows a single Load Balancer to act as a traffic cop, directing requests to different backend services based on the URL path, host headers, or query parameters.

1.1 Layer 7 (HTTP/S) Logic

Unlike Layer 4 load balancing (which only sees IP addresses and ports), Layer 7 load balancing inspects the actual HTTP request.

- **Path-based routing:** example.com/orders goes to the Order Service; example.com/users goes to the User Service.
- **Host-based routing:** api.example.com goes to the Backend; app.example.com goes to the Frontend.

1.2 URL Maps and Target Proxies

In GCP, the "URL Map" is the configuration object that defines these rules. It sits behind a Target Proxy and routes traffic to various **Backend Services**.

Section 2: Internal vs. External Service Exposure

Deciding how to expose a service is a critical architectural decision based on the intended consumer.

2.1 External Exposure (Public)

- **GCP External HTTP(S) Load Balancer:** The entry point for global users. It provides Anycast IP (one IP for the whole world), DDoS protection (Google Cloud Armor), and SSL termination.

- **Use Case:** Your public-facing web portal or mobile app API.

2.2 Internal Exposure (Private)

- **Internal HTTP(S) Load Balancer:** Resides entirely within your VPC. It has a private IP and is not reachable from the internet.
- **Use Case:** A "Payment Processing" service that should only be reachable by the "Checkout" service, never by a user's browser directly.

Section 3: Network Security Considerations

3.1 Google Cloud Armor

A Web Application Firewall (WAF) that integrates with the External Load Balancer. It can block traffic based on:

- **IP Allow/Deny lists:** Block known malicious IP ranges.
- **Geo-blocking:** Prevent traffic from specific countries.
- **SQLi/XSS filtering:** Protect against common web attacks.

3.2 Private Service Connect (PSC)

Allows you to consume services (like Managed Databases or Third-Party APIs) privately across VPCs without using external IPs or complex VPC peering.

Section 4: Professional Case Studies

Case Study: The "Blue-Green" Deployment Shift

Scenario: A large SaaS provider needs to update their "Reporting Engine" without a second of downtime.

Problem: A standard rolling update might cause errors if the new version is incompatible with the database for a few minutes.

Solution: They deployed two versions of the service behind an **Advanced Load Balancer**.

- They used the **URL Map** to keep 100% of traffic on "Blue" (v1).

- They sent "Green" (v2) traffic only to developers based on a specific cookie.
- Once validated, they performed a "Global Weighted Swap" in the Load Balancer, moving all users to Green instantly.

Section 5: Hands-On Lab - Advanced Routing & Security

This lab demonstrates setting up a single Load Balancer to route to two different microservices and securing it.

Phase 1: Create Backend Services (CLI)

Assume you have two GKE Service groups: video-service and image-service.

Bash

```
# 1. Create Health Checks
gcloud compute health-checks create http http-basic-check --port 80


# 2. Create Backend Services
gcloud compute backend-services create video-backend \
    --protocol=HTTP --port-name=http --health-checks=http-basic-check --global


gcloud compute backend-services create image-backend \
    --protocol=HTTP --port-name=http --health-checks=http-basic-check --global
```

Phase 2: Configure the URL Map (The "Traffic Cop")

This is where the multi-service routing logic lives.

Bash

```
# 1. Create the URL Map
gcloud compute url-maps create web-map --default-service video-backend
```

2. Add a path matcher to route /images to the image-backend

gcloud compute url-maps add-path-matcher web-map \

--default-service video-backend \

--path-matcher-name path-filter \

--backend-service-path-rules="/images/*=image-backend"

Phase 3: Secure with Cloud Armor (UI Steps)

1. **Navigate to Network Security:** Go to **Network Security > Cloud Armor**.
2. **Create Policy:** Name it block-malicious-traffic.
3. **Add Rule:** * **Action:** Deny.
    - **Match:** origin.region_code == 'CN' (Example: blocking traffic from a specific region) or use pre-configured rules like sqli-canary to block SQL injection.
4. **Apply to Target:** Select the web-map (Load Balancer) created in Phase 2.

Phase 4: Validating Secure Communication

To ensure the backend is truly secure and only accessible via the Load Balancer:

Bash

```
# Verify that the backend instances do NOT have public IPs
gcloud compute instances list --filter="name~'video-backend'"


# Test the routing
# This should hit the video service
curl http://[LB_EXTERNAL_IP]/home


# This should hit the image service
curl http://[LB_EXTERNAL_IP]/images/logo.png
```

Summary of Key Differences

Feature	Service (Type: LB)	Ingress / Advanced LB
Routing Level	Layer 4 (TCP)	Layer 7 (HTTP)
Intelligence	Simple Round Robin	Path/Host/Header based
Security	Firewall Rules only	Cloud Armor (WAF) + SSL
Cost	1 LB per service	1 LB for multiple services